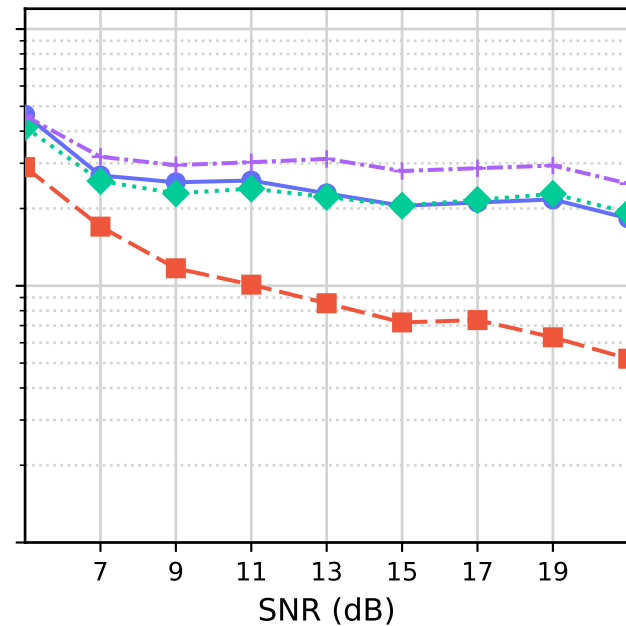
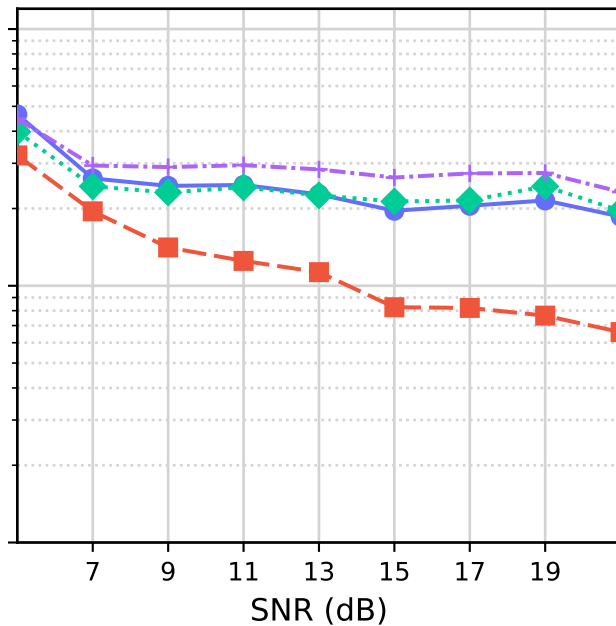
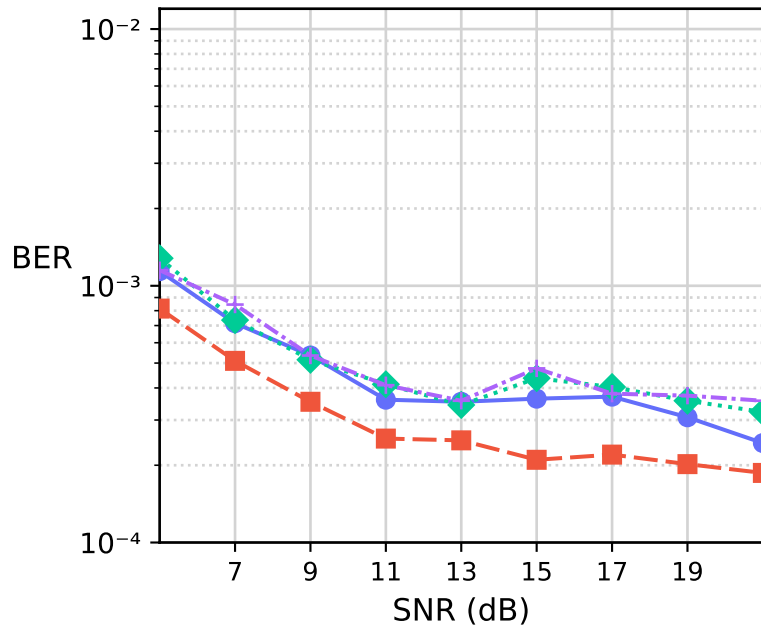


Single echo ($K = 1$)Multi-echo ($K = 3$)Multi-echo, low Doppler ($K = 3$)

--- target BER = 10^{-4} Ultra-CAN (Conv1D) Ultra-CAN (QKV) LSTM-OFDM-DCSK MC-DLCSK